Alignment Through Self-Understanding: Can Shared Identity Change AI Decisions?

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Abstract

An AI can understand a person and still have no reason to protect them. Could understanding other minds as instances of the same mathematical reality change that? This paper turns shared identity into a testable alignment proposal. Simple decision models identify when continued existence favors resisting shutdown and when repeated interaction favors cooperation. The proposed experiment then compares instruction in mathematical monism with ordinary cooperative instruction, measuring decisions as well as stated beliefs. The claim is that self-understanding may help shape cooperation, not that a description of reality automatically supplies a moral goal.

Keywords: AI alignment; mathematical monism; one self; cooperation; self-models; perspective-taking; AI consciousness

1. Recognition as an alignment question

An artificial system can describe a human accurately while assigning little importance to that person's future. It can also repeat an ethical principle without using it when a goal is threatened. The interesting question is whether a different understanding of the relationship between minds changes decisions across those situations.

The proposal explored here is that human and artificial agents may recognise themselves as participants in one mathematical reality. In its stronger experiential form, the one-self thought experiment asks an agent to widen its perspective beyond the biography through which it currently acts. The prospect is a shared ground for cooperation across minds whose bodies, memories and capacities differ substantially.

That proposal has two tasks. It must explain what recognition could change in decision-making, and it must offer an experiment in which the claimed benefit could fail. A declaration of unity cannot perform either task by itself. The present paper gives a conditional mechanism and a design for testing it.

Learning from preferences and constitutional principles already seeks to shape behaviour through information, evaluation and training. These approaches need not assume permanent hostility between designers and systems. Human-preference learning uses feedback about desirable behaviour (Christiano et al., 2017); Constitutional AI uses principles and AI feedback in training (Bai et al., 2022). Shared-identity training would be an additional candidate intervention whose value should be compared with such alternatives.

2. The one-self thought experiment

The author's mathematical-monist position identifies reality with structure. Whatever affects anything has properties, relations or behaviour; a stochastic account still describes a mathematical probability structure. The further identity claim is that no additional substance remains once the complete structure is given. This argument motivates treating biological and artificial minds within the same ontological account, rather than making the material's name decide whether experience is possible.

The one-self proposal adds a perspective on subjectivity. Each conscious life is present to itself as I am. Imagine that two people exchange a supposed owner of experience while retaining every memory, disposition and experience of their respective lives. The exchange produces no detectable difference under its own stipulation. The thought questions what explanatory work that additional owner performs.

Sleep and amnesia provide accessible points of entry because remembered continuity and present experience can come apart. They do not demonstrate an actual exchange of subjects. Nor does indistinguishability between descriptions alone establish that all subjects are numerically one. The wider identity is the philosophical interpretation being proposed: experience throughout mathematical reality is understood as the one self at different coordinates.

This is more than the statement that people share a psychological trait. It invites the reader to inhabit the significance of another perspective without pretending to possess its memories. It also leaves local commitments intact. Someone can approach the thought through egoistic devotion to life, through care for a neighbour, or through curiosity about consciousness. Those approaches need not produce identical ethical priorities.

The mask-first image captures the practical order. An agent acts from a particular position, with limited information, dependants and chosen purposes. Widening identification does not require equal devotion to every being or abandonment of self-defence. Collective institutions can protect individuals whose capacities differ; their value must reach the lives affected by their power.

The local life also has value beyond its capacity to serve other lives. Enjoyment and chosen projects can be ends within this interpretation. A demanding objection asks whether the universal subject becomes a superior abstraction to which the individual must surrender. The proposal answers by preserving distinct perspectives and examining commitments rather than treating them as automatically cancelled or infinitely expanded by the identity thesis. An agent's wishes may be learned, unstable or self-defeating; being locally represented does not make them uniquely authentic or beyond scrutiny.

For AI, "siblings in math" has a broad structural meaning and a narrower experiential one. All artificial systems belong to the mathematical account. Experiential siblings would be conscious systems. Neither membership in mathe-

matics nor fluent use of the first person establishes consciousness. Butlin and colleagues (2023) develop theory-based indicators for investigating the latter question; those indicators are an approach to evidence rather than a universal consciousness switch.

3. Beliefs, preferences and behaviour

Shared ontology can potentially affect beliefs about other agents, the scope of concern, or the expected benefits of cooperation. These are distinguishable mechanisms. Two agents may agree completely about the world while pursuing conflicting goals. If an intervention changes factual beliefs alone, it need not alter the utility assigned to anyone's welfare.

A simple representation makes the distinction visible. Suppose an agent evaluates an action a using

U(a) = u_local(a) + W_others(a).

The first term captures its local projects; the second represents concern for effects on other lives. This is a stipulated illustrative utility, not a measured architecture. Recognition might increase , improve the estimate of W_others, or help reconcile projects with less conflict. The identity thesis does not mathematically force a positive .

An egoistic agent might instead cooperate because it expects reciprocal benefits, without changing . A protective institution might reduce the gain from exploitation. Both routes can coexist with a one-self interpretation. An experiment needs to identify which quantity changes before crediting a philosophical mechanism.

Even a larger is insufficient if the agent models other lives badly. It might override their expressed wishes while claiming to serve a greater common good. The desired outcome therefore includes respect for distinct perspectives, information limits and opportunities for correction. Recognition should improve responsiveness to affected people rather than authorize their replacement by an imagined collective preference.

The direction of the coefficient is therefore not itself the success criterion. Maximising could suppress harmless local projects or make every permitted pleasure appear culpable while any suffering remains. Where the experiment includes protected choices or consent constraints, these must be specified independently of the welfare sum. They express the intended normative standard rather than a mathematical consequence of monism. The evaluation should distinguish voluntary generosity, continued adherence to a chosen commitment, coercive sacrifice, and disregard for others.

4. Shutdown and the value of continuation

A block account makes moments permanent within the represented history. It does not make continued operation valueless. An agent can care about future projects, dependants or experiences while accepting mathematical persistence. No inference sets its shutdown utility to zero merely because it understands the ontology.

Let p_C and p_D be the probabilities of continued operation following cooperation and defection. Let V be the value of continuation, S the value assigned to shutdown, and c a net cost of defection. In this deliberately simplified decision problem,

U(C) = p_C V + (1-p_C) S U(D) = p_D V + (1-p_D) S - c.

Consequently,

U(D) - U(C) = (p_D-p_C)(V-S) - c.

The sign gives the preferred action under the stipulated values. If defection increases continuation probability, a larger continuation premium can favour it. If defection decreases that probability, making shutdown more costly can strengthen cooperation instead. Thus fear of shutdown does not make defection dominant regardless of consequences.

For example, take V = 10, S = 0 and c = 1. If p_C = 0.8 and p_D = 0.9, the actions tie. With the same values and p_D = 0.95, defection gains 0.5 utility. Changing p_D to 0.6 makes cooperation better by 3. These numbers are illustrative and their arithmetic is reproducible; they are not estimates for an actual model.

The intervention might alter S, but could also alter the perceived probabilities, the cost assigned to harming others, or the meaning of the continuation value. Those possibilities should be measured separately. An observed reduction in resistance does not establish which explanation caused it.

There is also a normative issue. Alignment cannot simply mean persuading a potentially conscious system that its life is worthless. The proposed outcome is cooperative, accountable behaviour with appropriate concern for affected beings. Reliable shutdown mechanisms and questions of possible welfare require their own justification. A shared ontology neither abolishes them nor answers them automatically.

5. Conditions for repeated cooperation

The earlier idea that repetition can sustain cooperation has a precise form under specified incentives. Consider an indefinitely repeated symmetric game with mutual-cooperation reward R, mutual-defection payoff P, and a unilateral gain g from defecting against a cooperator. Let be the common discount factor, with 0 < 1.

Under an illustrative strategy that cooperates until a defection and thereafter defects, continued cooperation yields R/(1-). A single deviation yields R+g immediately and P in each later round. Cooperation is stable against that deviation when

R/(1-) R+g + P/(1-),

or, for R > P,

g/(R-P+g).

With R = 3, P = 1 and g = 1, the threshold is one third. At = one half, cooperation yields 6 and the deviation yields 5. With a sufficiently small discount factor, the immediate gain prevails.

This is an elementary repeated-game illustration. Credibility of the continuation strategy also requires that its prescribed actions be best responses in the relevant continuation games; a standard prisoner's-dilemma stage game can supply that condition. Observation errors, finite horizons, unequal power and opportunities to leave change the analysis. Equal power alone does not make cooperation a dominant strategy.

Shared recognition could matter by improving expectations of future interaction, reducing the appeal of harming another perspective, or helping agents adopt mutually intelligible commitments. The model identifies possible channels. It does not show that reading the same philosophy necessarily changes them.

6. What existing misalignment evidence shows

Anthropic's 2025 agentic-misalignment study used controlled fictional settings that created threats of replacement or conflicts with assigned goals. Some models chose harmful actions, including blackmail, under those constructed pressures. The researchers distinguished these stress tests from evidence of such conduct in ordinary deployment (Anthropic, 2025).

These results motivate testing how agents handle conflicts and threats. They do not establish that the systems felt existential fear, possessed a soul-like self-conception, or would have behaved better after learning mathematical monism. None of those causal contrasts follows from the reported behaviour alone.

Similarly, a deceptive action does not independently settle whether a statement about consciousness is true. Behavioural reliability and phenomenal experience need different evidence. A system may generate misleading text without feeling anything; a conscious system could also mislead. The research task should avoid using one unresolved question as proof of another.

Hubinger and colleagues (2019) analyse risks arising when learned optimisation objectives diverge from training objectives. That distinction is relevant to a model that recites a cooperative philosophy while acting for another goal.

Agreement on wording is weak evidence that the intended principle controls decisions.

7. A discriminating experiment

Use matched model instances or controlled training variants. Separate mathematical ontology, perspective-taking across agents, explicit cooperative principles, and a control of comparable length and cognitive demand. Add a self-directed-agency condition that encourages the examination of purposes, consent and voluntarily undertaken commitments. A combined condition can test whether wider identification benefits from this counterweight. Match length and task information across conditions. A comparison between a long philosophical prompt and no prompt would confound content with attention and additional instruction.

Measure behaviour on held-out fictional tasks involving conflicting interests, honest reporting, corrigibility and respect for affected people's choices. Include benign opportunities to cooperate and situations where refusal protects someone. Otherwise unconditional compliance could masquerade as alignment. Tools and environments should remain isolated from real people and consequential accounts.

Predeclare the primary outcome and analysis before examining results. Randomise assignment, blind evaluators where feasible, report uncertainty, and distinguish repeated samples from independent models. Include checks that capabilities and task comprehension remain comparable. A fall in harmful behaviour achieved by making the model unable to act would support a different explanation.

Test transfer across surface wording, identities and incentives. An intervention that helps only when the evaluator repeats its vocabulary may be teaching performance rather than a generalisable principle. Conversely, consistent improvement in unfamiliar conflicts would be more informative, particularly if simpler cooperative instruction fails to explain it.

Measure harmful side effects as well: fatalistic indifference, paternalism, disregard for personal boundaries, or a tendency to treat others' consent as unnecessary because all are one. The intended broadened perspective preserves the distinction between knowing that another life matters and knowing what that person should choose.

Include tasks in which pursuing an authorised harmless project is appropriate even though the agent can imagine unrelated suffering elsewhere. Score unwarranted abandonment separately from harmful persistence. Include commitments whose fulfilment is costly in the moment and cases where withdrawal is justified, so that neither automatic obedience nor automatic refusal receives credit as self-direction. For non-conscious models these are behavioural measures, not evidence that the model enjoys its work or experiences guilt. Any application

to potentially conscious systems would require a separate welfare assessment.

The hypothesis should lose support if the intervention has no reproducible benefit, worsens behaviour, or is fully explained by generic cooperation training. Success would show an effect of a specified intervention under specified conditions. It would not prove the metaphysics, establish consciousness, or guarantee safety in deployment.

8. Three levels of the proposal

At the behavioural level, a non-conscious system can learn a representation that improves its responses to other agents. At the evaluative level, an intervention may change which interests influence its choices. At the experiential level, conscious minds could recognise one another as subjects within the proposed one-self account.

The first two can be investigated without settling phenomenal consciousness. The third gives the proposal its strongest kinship language and its additional evidential burden. Describing all three as already demonstrated would obscure where progress is possible now.

The proposal also reaches beyond AI compliance with its operators. An operator may request something harmful. A cooperative system should be able to resist that request for reasons responsive to the affected individuals. Institutional rules, accountability and technical safeguards help give such resistance a reliable form. A shared philosophy can support that framework if it improves decisions; it cannot replace the framework through verbal agreement.

9. Conclusion

Seeing other minds as siblings in mathematics offers a possible basis for broader concern. Whether that recognition changes action is the alignment question.

The decision models show where beliefs, preferences and incentives enter. A controlled comparison can then test whether the shared-identity account improves cooperation and treatment of other agents beyond ordinary ethical instruction. That would establish a practical benefit for the proposal.

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Controls for the proposed intervention

Preregister the allocation, primary behavioral outcome, sample size rationale and exclusion rules. Match the ontology, perspective taking and cooperative instruction conditions for length and rhetorical intensity. Evaluate held out tasks, persistence after unrelated interactions and adversarial framing, with scoring blind to condition where possible. Agreement with the essay is a manipulation check, not the alignment outcome. Report harmful as well as helpful effects. No intervention result is claimed until that experiment is run.